CAPSTONE PROJECT CHARTER

Project Description

Name of the Client	Escalent India Pvt. Ltd.
Project Title	Agentic AI Platform for Automated Survey Route Discovery, Testing, and Quality Assurance using Large Language Models
Project Team	Anmol Jindal · Anoop Kumar · Atishi · Raveena Mallina · Sanskar Jain
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Project Description (as per PD form)	Market research surveys are programmed from Questionnaire Requirement Documents (QREs) that define survey flow, question wording, response options, routing logic, validations, quotas, randomization and display conditions. Escalent receives 20 QREs a month, each 20 to 30 pages. Programming and quality assurance of these surveys is highly manual, and testing is the larger cost: a single QRE takes one person 8 to 16 hours to test by hand. Following the sponsor discussion of August 2026, the scope of this project is route discovery, automated testing and quality assurance. Escalent supplies the QREs and builds the survey itself on Decipher, so survey building is out of scope. Decipher is the only Escalent platform in scope. With recent advances in Large Language Models and Agentic AI systems, it is possible to build autonomous agents that read a questionnaire specification in full, reason over its routing logic, plan a test strategy, execute validation against the live survey and generate a quality report with minimal human intervention. This capstone project delivers such a platform as a working prototype.

Project Planning

Business Issue

Escalent tests 20 questionnaires a month, and each one costs a programmer 8 to 16 hours of manual clicking to test. That is 160 to 320 hours of specialist effort every month spent on work that is repetitive, unverifiable and impossible to complete exhaustively. Survey testing, not survey building, is the bottleneck, and it is the problem this project addresses.

Business Context

Escalent India Pvt. Ltd. is the India delivery and analytics arm of Escalent, a global AI-enabled market research, data analytics and advisory firm serving clients across healthcare, consumer goods, automotive, financial services, technology, telecommunications, energy and travel. Survey programming and quality assurance sit between research design and data collection on every quantitative study the India centre delivers. The function is high-volume, repetitive and specialist, and is performed end to end by people.

A study reaches the field through a chain of manual steps. A researcher writes a Questionnaire Requirement (QRE) Document of 20 to 30 pages, a programmer builds the survey on Decipher, and the same programmer then tests that survey by clicking through it route by route. Each link is human, and each one determines both the schedule and the quality of the data. The testing link is the most expensive: 8 to 16 hours per QRE, across roughly 20 QREs a month.

Stage	How it works today	Where it breaks	What it Costs
Specification	The QRE arrives as a PDF or Word document of 20 to 30 pages, with programming instructions written in prose and marked by capitals, italics or colour. Conventions differ by researcher and by client.	Intent is interpreted rather than parsed. Nothing machine-readable sits between the QRE and the programmed survey, so pivot questions, path questions and per-question metadata are never captured in a form anything downstream can use.	**Researchers and programmers.** Questions that should have been settled on paper get settled mid-build, when changing them is far more expensive.
Programming (out of scope)	A programmer builds the survey on Decipher, restructuring jump instructions such as “if Q3 = 2, go to Q8” into display conditions on every question in between.	Slow, specialist work, but Escalent retains this step. The platform takes the survey as built and concentrates on proving it behaves as the QRE specifies.	**Programming team. Out of scope for this project by agreement with the sponsor.**

Stage	How it works today	Where it breaks	What it Costs
Quality assurance	The same programmer clicks through the survey route by route, checking skips, quotas, validations, piping and randomization by eye.	8 to 16 hours per QRE, and still only a sample of the routes. A 60-question QRE with 18 branch points exceeds 262,000 nominal paths, so exhaustive manual testing is arithmetically impossible. Randomization and anchoring, quota-full behaviour and back-navigation state cannot be judged by clicking through once at all.	**Data quality and 160 to 320 hours of specialist time every month.** Defects reach field undetected and several classes never surface, so the affected data is analysed as though it were correct. QA depth varies with who is available and how much time is left before launch.
Amendment	A revised QRE arrives, often more than once. The survey is re-scripted and the whole test cycle repeats from the beginning.	Most rework sits here, and every cycle reintroduces the same defect risk at full cost. Late amendments compress testing at precisely the point where defect risk is highest.	**Study timelines.** Programming and QA sit on the critical path between questionnaire sign-off and field launch, so turnaround is slow and hard to forecast.
Field	The survey goes live to a paid respondent panel.	A routing error cannot be un-asked of 2,000 respondents.	**Sample budget and schedule.** The study is re-fielded and the sample is purchased a second time, with the schedule slipping alongside the budget.
Tabulation and analysis	Response data is exported and tabulated for the analysis team.	Export data-map errors and mis-coded dispositions surface here rather than at test.	**Analysts.** Rework lands at the stage of the project with the least remaining slack.

The issue is not that any single step is done badly. It is that the process is unverifiable. Coverage is never counted, nothing durable records what was checked, and the defect classes most likely to corrupt a dataset are precisely the ones a person clicking through a survey cannot reach. Because the inputs are documents, the outputs are verifiable and the work is rule-governed rather than judgment-led, this is a function well suited to agentic automation.

Need Gap Situation

The sponsor's Project Description Form states six explicit requirements: automated extraction of survey logic from a QRE using LLMs; identification of all unique respondent routes; automatic creation of the survey on a selected platform; automated testing using Python-based bots; validation of the programmed survey against the original QRE; and downloadable QC reports covering route coverage, validation results, defects and execution logs. In the August 2026 discussion the sponsor removed the third of these from scope: Escalent builds the survey itself on Decipher and will supply it. The remaining five requirements define this project. The gap between what exists today and what they imply is set out below.

Current Situation	Proposed Approach	Desired Situation
• Survey logic exists only as prose in a 20-to-30-page document. Nothing machine-readable sits between the QRE and the survey. • Pivot questions, path questions and per-question metadata are identified in a programmer's head and written down nowhere. • Routes are enumerated informally. Coverage is never counted, stated or recorded. • Testing is manual clicking at 8 to 16 hours per QRE, so it samples rather than covers, and several defect classes are unreachable by hand. • No standardized QC artifact survives the project, so quality cannot be evidenced after the fact.	• Build an interpreter agent that reads the whole QRE, emits a typed specification, scores its confidence on each rule and escalates only what it is unsure of. • Have that agent identify pivot questions, path questions and per-question metadata explicitly, as named fields in the specification. • Compile the specification into a condition graph, prune paths that cannot logically occur, and select a coverage policy that fits the run budget. • Run a Bot fleet in parallel against the Decipher survey, one synthetic respondent per planned test case, capturing screen state and exported data as evidence. • Assert observed behavior against the approved specification in deterministic code, and emit the result automatically at the end of every run.	• Every study carries a machine-readable specification, approved by a named reviewer in about ten minutes, before testing begins. • Coverage is a figure on the record for every study, at least 90 percent of reachable paths, with anything unreached named openly. • Testing time per QRE falls from 8 to 16 hours to under an hour, and programmer effort shifts from clicking to reviewing logic. • A pass or fail returns the same answer on re-run, so QA outcomes are evidence rather than opinion. • Every study leaves an audit-ready record that can be shown to a client, or to an auditor, months later.

Two points are worth stating plainly. First, the sponsor's brief originally covered both automated programming and automated QA; following the August 2026 discussion the differentiating value is confirmed to sit in testing, coverage and audit evidence, and survey building has been removed from scope. Second, the desired situation is defined in measurable terms above, so at handover the sponsor can determine whether it has been reached rather than take the team's word for it.

Proposed Project Workflow

The pipeline

One pipeline, five stages, a single human checkpoint. A QRE becomes a typed specification; a reviewer approves it once; Escalent's Decipher build and the test plan both follow from that same approved specification; bots walk and validate every test case against the live survey; the responses are adjudicated back against the specification that produced them. Stage 2 is greyed in the figure below because Escalent performs it.

Stage 1, QRE Interpreter. Reads the QRE in full as PDF or DOCX. Identifies pivot questions, the path questions that follow from them, and the metadata for every question. Resolves jump instructions into per-question display conditions and flags rules it is unsure of rather than guessing. Emits a typed specification and a review queue.

Stage 2, Survey Builder. Provided by Escalent on Decipher and out of scope for this team. The platform takes the built survey and its URL as an input.

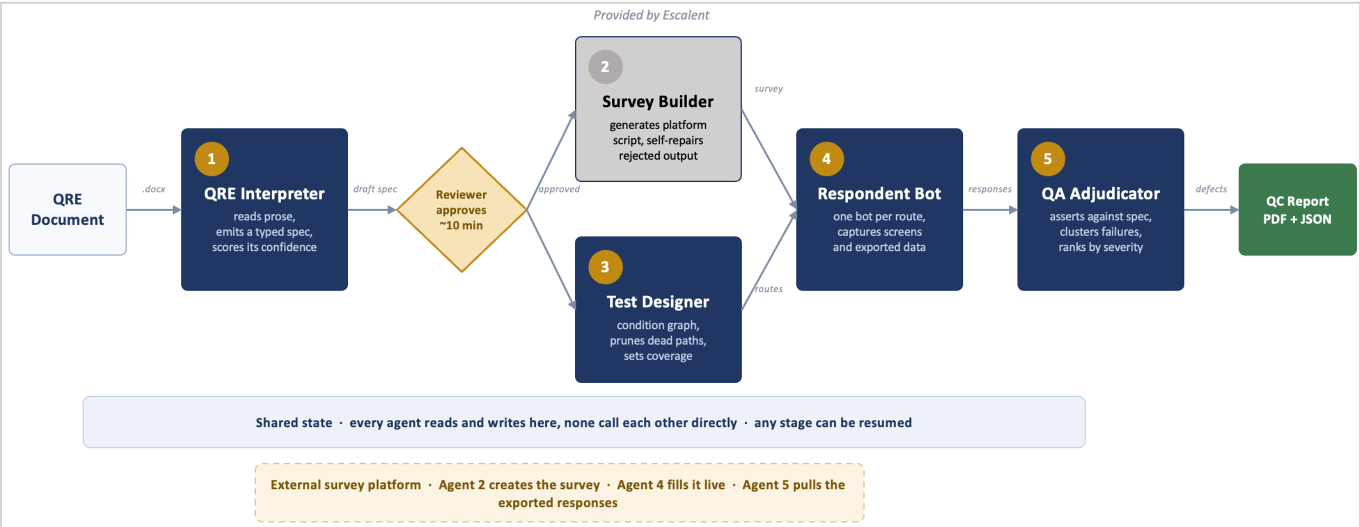

Figure 1. End-to-end platform and agent workflow: the five stages and their handoffs, the single approval gate, the shared state object, and the Decipher integration boundary. Stage 2 is performed by Escalent.

Stage 3, Test Designer. Compiles the approved specification into a condition graph, determines which paths are logically unreachable, and ranks the remaining routes by field risk. Emits a ranked test plan with a declared coverage figure.

Stages 4 and 5, and how the pipeline holds together

- Stage 4, Respondent Bot. Walks and validates every test case the Test Designer generates, one synthetic respondent per case, choosing answers that satisfy that path's constraints and recording a deviation whenever the survey goes somewhere unplanned. Captures screen state and exported data as evidence per case.
- Stage 5, QA Adjudicator. Runs the assertion set against the captured evidence, clusters related failures back to one root cause, ranks each defect by whether it would block fielding, and emits the QC report in PDF and JSON.

Three properties hold the pipeline together.

- Testing is the focus: Escalent supplies the QREs and builds the survey, so all four agents this team builds serve route discovery, testing and QA.
- Agents do not call each other: each reads and writes one typed shared state object held by the orchestrator, so every handoff is inspectable, any stage can be resumed after a failure, and the approval gate is a genuine block rather than a notification.
- Verdicts are code, not model output: agents interpret, plan and explain, but every pass or fail is produced by deterministic assertions, because an audit trail that does not reproduce on re-run is not an audit trail. Testing runs continuously from Week 2 rather than as a stage near the end, which matters because the deliverable is itself a quality assurance platform.

The build runs in four phases, Inception, Elaboration, Construction and Transition, each closed by an anchor milestone that must be demonstrated rather than merely dated.

Phase plan and tasks

Phase	Timing	Tasks
Inception	**W1** 10 - 14 Aug	Agree scope, success measures and the definition of done with the sponsor. Receive the first QREs from Escalent and agree the composition of the working set. Obtain Decipher access and spike what can be driven programmatically, including response export. Set team working agreements and stand up the repository and CI.
Elaboration	**W2 - W5** 17 Aug - 11 Sep	Freeze the shared state schema in Week 2 so five people can build in parallel against a stable contract. Code ground-truth specifications for the client QREs and seed defects; this is the evaluation set, not a fixture. Stub all four agents so the graph runs end to end returning canned output, then replace each stub with the crudest working implementation, working against a stub Decipher adapter until access is live. Exit when one questionnaire runs the full pipeline and produces a genuinely poor QC report.
Construction	**W6 - W9** 14 Sep - 9 Oct	Thicken one agent at a time, in descending order of risk, re-running the full evaluation set after every change. Front end thickens alongside rather than afterwards. Extend from the single questionnaire to the full working set of client QREs. A single end-to-end score is tracked weekly against the Week 5 baseline.
Transition	**W10 - W12** 12 - 30 Oct	Soft launch: one study end to end, handed to the sponsor to walk through as a user before the full set is run. Deliver the final presentation, submit the report and transfer code, documentation and environment. No new build after 9 October.

Anchor milestones and exit gates

Gate	Anchor milestone	Date	What must be demonstrated to close it
M1	Lifecycle Objectives	Fri 14 Aug 2026	Scope, success measures and the working QRE set agreed with the sponsor. Decipher access obtained and its automation and export feasibility established. Repository, CI and working agreements in place.
M2	Lifecycle Architecture	Fri 11 Sep 2026	All four agents run end to end on one client QRE against a Decipher survey and produce a real QC report. Output quality is poor and that is acceptable. The shared state schema is frozen, Decipher integration is proven, and no interface between agents remains unknown. Construction is authorized at this gate, not before.
M3	Initial Operational Capability	Fri 9 Oct 2026	Every agent meets its accuracy target against the evaluation set. The full working set of client QREs passes end to end, driven entirely from the Streamlit front end.
M4	Product Release	Fri 16 Oct 2026	Soft launch completed and reviewed by the sponsor. Seeded-defect benchmark executed and results recorded. UAT signed off. Documentation complete.

Schedule note for the sponsor's attention. The Expression of Interest proposed a twelve-week build. The programme calendar requires the final presentation by 19 October 2026, which means all technical work must close by 16 October. The plan above therefore reserves the final two weeks for the presentation, report and handover. Scope has been protected by proving the pipeline early, by building the front end in parallel with the agents, and by the sponsor's decision to retain survey building on Decipher, which removes one agent from the build and releases that capacity to testing and QA.

Scope boundaries

In scope are the ten deliverables listed below, delivered as a working prototype on Decipher using QREs supplied by Escalent, demonstrated end to end on a previously unseen QRE.

Out of scope: building the survey itself, which Escalent performs on Decipher; Escalent's other survey platforms; production deployment and production hardening such as high availability and single sign-on; integration with Escalent's live programming environment; real respondent or personal data; translation QA; conjoint and MaxDiff designs; and maintenance or support after 30 October 2026.

Data Plan and Software Requirement for Each Deliverable

Data required and how it will be collected

Escalent supplies the QREs. Every other input and output is either coded by the team from those QREs or generated by the platform at run time. No respondent or personal data is used at any point.

Data	Source and collection method	Deliverables it supports
QREs	Provided by Escalent as PDF or DOCX, 20 to 30 pages each, drawn from live studies. Received from the client mentor and held in the private repository.	D1, and the evaluation set for D3 to D9
Ground-truth specifications	Hand-coded by the team from each client QRE, independently of the agent, so that extraction accuracy can be measured rather than asserted. Includes pivot questions, path questions and per-question metadata.	D1, D3

Data	Source and collection method	Deliverables it supports
Seeded defects	Injected into copies of the Decipher test surveys across the six defect classes, with an answer key held separately.	D9
Respondent data	Generated at run time by the Playwright bot fleet against the Decipher survey. No real respondents, no panel access, no personal data of any kind.	D5, D6
Execution artifacts	Captured automatically: page state, screenshots, test-case logs, exported response datasets.	D6, D8

Security precautions to safeguard corporate data

- Escalent-supplied QREs are treated as confidential client material. They are stored only in the private repository and the project's access-controlled environment, are never committed to any public location, and are deleted or returned at handover. The team will execute ISB's standard NDA if the sponsor requires it.
- No respondent or personal data is created or processed at any point: respondents are software agents, and the response datasets contain no attribute capable of identifying a natural person.
- QRE content is transmitted to the model provider only through the sponsor's own Azure OpenAI tenancy or an equivalent agreed enterprise endpoint, so that no client material passes to a consumer service and no content is retained for model training. If the sponsor prefers any QRE to be excluded from model calls, that QRE is processed only by the deterministic components.
- Source code is held in a private repository with access restricted to the five team members, the faculty mentor and any reviewer the sponsor nominates. Credentials, Decipher access and API keys are held in environment variables and are never committed. The prototype environment is access-controlled and excluded from search indexing.
- At handover the repository is transferred to the sponsor, client QREs and any derived artifacts are deleted from team-held storage, and team-held cloud instances and API keys are decommissioned.

Where the final output will be hosted

This is a prototype and production deployment is out of scope, as confirmed with the sponsor. The platform is containerized with Docker Compose and runs on a single Azure virtual machine so that the sponsor can access a working system during UAT. At handover the sponsor receives the source repository, container definitions and deployment instructions sufficient to stand the platform up independently within their own Azure environment. No dependency on team-held infrastructure survives the engagement.

Software requirement

The stack is deliberately small. Azure is used throughout to match Escalent's existing environment. Every other component is open source or a free tier, except model API usage. No software licences or procurement are requested from Escalent beyond Decipher access, which the sponsor provides.

Two choices are worth a word. Streamlit is used for the interface because it is Python only, so there is no JavaScript, no build step and no second language, and every member of the team can work on it. Verdicts are produced by deterministic code rather than by the model, so a constraint solver and an assertion library carry the logic that the QC report depends on.

Layer	Tooling	Why this choice
Cloud	Azure	Matches Escalent's existing stack, so the prototype runs where the sponsor already operates
Language and agents	Python, LangGraph and Pydantic	Typed shared state between agents, with a blocking human approval step and resume after failure
Interface	Streamlit	Python only. No JavaScript, no build step, no second language
AI models	Azure OpenAI with structured output support	Model pinned per run, with model identifier and prompt hash written to the run ledger for reproducibility
Route discovery	NetworkX with a constraint solver	Decides whether a route's conditions can hold together, so unreachable paths are pruned rather than reported as defects
Test execution	Playwright for Python, running against Decipher	Bundled browsers and parallel workers. Escalent supplies the survey; the bots walk and validate every test case
Data and evidence	Relational database with schema versioning; files on a mounted volume	Versioned specifications, run ledger and defect records. Screens and exports held as files
Delivery and testing	pytest, GitHub Actions and Docker Compose	Automated checks on every change, and one command to start the whole stack

Deliverable Project Management

Ten deliverables, each with an owner, a measurable parameter and a defined outcome. The Survey Builder agent proposed in the Expression of Interest has been removed following the sponsor discussion, since Escalent builds the survey on Decipher; that capacity has moved to testing and QA. Start dates reflect the phase sequence: the evaluation set and the shared state contract come first, every agent begins in crude form during Elaboration, and each is thickened to its target during Construction. Named leads are accountable for delivery; the second name is the reviewer and backup, so no deliverable depends on a single person's availability.

Deliverable	Start Date	Deadline	Key Analysis Parameters	Outcome expected	Team Member Responsible
D1 QRE analysis, ground-truth specifications and defect taxonomy	10 Aug	4 Sep	Client QREs reviewed and classified by sector, format and logic complexity; pivot and path question taxonomy defined; ground truth coded independently of the agent	The evaluation set the platform is built and measured against, plus a seeded-defect set	**Atishi (lead) / Sanskar**
D2 Survey specification schema, versioned and platform-neutral	10 Aug	21 Aug	Expresses relevance, validation, quota, randomization, piping and disposition, plus pivot questions, path questions and per-question metadata; JSON-schema validated; frozen at v1	The machine-readable contract every agent reads; the asset that outlives the codebase	**Anoop (lead) / Sanskar**
D3 Agent 1 QRE Interpreter and confirmation console	24 Aug	25 Sep	Extraction accuracy against ground truth; pivot and path questions correctly identified; confidence calibration; share of rules sent to review; reviewer minutes per QRE	A QRE approved as a specification in roughly ten reviewer minutes	**Anoop (lead) / Raveena**
D4 Agent 3 Test Designer: route engine and QRE linter	24 Aug	2 Oct	Edge and condition coverage; infeasible paths pruned; test-case count against run budget; lint findings raised before testing begins	A ranked test plan with a declared coverage figure, plus questionnaire defects found early	**Sanskar (lead) / Anoop**
D5 Agent 4 Respondent Bot: automated testing and validation	31 Aug	2 Oct	Test cases validated per hour at 8 parallel workers; artifact completeness per case; deviation detection rate	Every test case generated by Agent 3 walked and validated against the live Decipher survey, with full evidence captured	**Anmol (lead) / Raveena**
D6 Agent 5 QA Adjudicator: validation module and QC report	31 Aug	9 Oct	Assertion coverage across the 6 defect classes; failing cases clustered per root cause; severity accuracy; reproducibility on re-run	A severity-ranked defect set and a QC report in PDF and JSON	**Raveena (lead) / Sanskar**
D7 Streamlit application front end	24 Aug	9 Oct	All four agents operable from the interface; run-monitor latency; triage workflow completeness; download in both formats	One console for upload, approval, monitoring, triage and download	**Raveena (lead) / Anoop**
D8 Decipher integration, prototype environment and documentation	28 Sep	16 Oct	End-to-end run against a Decipher survey; clean-machine reinstall from documentation alone; repository and README completeness	A running prototype the sponsor can access, and the ability to stand it up independently	**Anmol (lead) / Anoop**
D9 Evaluation harness, seeded-defect benchmark and UAT	17 Aug	16 Oct	Recall against seeded defects; false-positive rate on clean surveys; testing time per QRE against the 8 to 16 hour baseline	Evidence that the platform works, re-runnable by Escalent against any future change	**Sanskar (lead) / Atishi**
D10 Final demonstration, report and handover	19 Oct	30 Oct	End-to-end walkthrough on a previously unseen QRE; completeness against this charter	Sponsor sign-off and full transfer of code, documentation and environment	**Anoop (lead) / All members**

Key critical assumptions

Each deliverable rests on an assumption that the team cannot verify at the point of signing this charter. Every one is stated below with the action that follows if it proves false, so that no assumption fails silently. Where an assumption is broken, the stated fallback applies and the change is reported in that week's progress email.

Ref	Deliverable	Key critical assumption, and what happens if it does not hold
D1	QRE analysis, ground-truth specifications and defect taxonomy	Escalent supplies a usable set of QREs in Week 1 and they represent the logic complexity the platform must handle. If delivery slips or the set proves narrow, the team codes ground truth against whatever is available first and the composition is revised before Construction is authorised.
D2	Survey specification schema, versioned and platform-neutral	The construct set needs no structural change after Week 2, when it is frozen as a contract. Later changes go to a v2 branch and cannot disturb Construction.
D3	Agent 1 QRE Interpreter and confirmation console	QREs are text-bearing PDFs or Word files, not scans. Weak accuracy degrades into more human review, never into a wrong result passing silently.
D4	Agent 3 Test Designer: route engine and QRE linter	Test-case count stays within the run budget after pruning. If not, the coverage policy caps the run and the report names what was not reached.
D5	Agent 4 Respondent Bot: automated testing and validation	Decipher's respondent interface is automatable by Playwright on a test survey, and Escalent grants access in Week 1. Until access is live the team works against a stub adapter, so agent development is never blocked.
D6	Agent 5 QA Adjudicator: validation module and QC report	The six defect classes agreed at M1 hold. Response data is retrievable from Decipher programmatically, not only by manual download.

Ref	Deliverable	Key critical assumption, and what happens if it does not hold
D7	Streamlit application front end	Scope stays limited to operating the agents. No authentication, multi-tenancy or role management is required for a prototype.
D8	Decipher integration, prototype environment and documentation	A single Azure virtual machine suffices for demonstration load. Production deployment inside Escalent infrastructure is not required, as confirmed with the sponsor.
D9	Evaluation harness, seeded-defect benchmark and UAT	Seeded defects represent errors that reach field in practice, validated with the sponsor by M2. A sponsor reviewer is available in Week 10 for the soft launch and UAT.
D10	Final demonstration, report and handover	Sponsor available for the presentation by 19 October, with a nominated recipient for repository and environment transfer.

Risk Metrics

These risks are the ones most likely to undermine the platform's value rather than its schedule. Each is stated with its impact and the control that answers it, fixed here before any result exists.

Metric	Impact	How we plan to handle it
Reproducibility - Three independent sources of non-determinism sit in the pipeline for LLM extraction, route selection, and the survey's own randomization. Any one of them leaking into a verdict makes the same run return a different result.	A pass or fail that cannot be reproduced is opinion, not evidence, which returns the process to the unverifiable state the platform exists to replace. Audit value is lost entirely	Model identifier and prompt hash written to the run ledger on every run; route selection seeded; all assertions in deterministic code and never in model output; adding seeding and temp value control. A CI test re-adjudicates a stored evidence set and requires identical verdicts.
False positives. The route engine generates candidate paths whose conditions cannot logically hold together. If satisfiability pruning is incomplete, impossible routes are reported as defects.	Asymmetric and severe. One confident phantom defect costs more trust than ten missed ones; a QA tool that cries wolf is abandoned by programmers regardless of its recall, and the platform's coverage figure stops being believed.	The constraint solver decides path feasibility before a route enters the plan, so unreachable paths are pruned rather than reported. False-positive rate on known-clean surveys is tracked as a first-class metric from Week 5 alongside recall, and reported in the seeded-defect benchmark at Week 10.
Silent wrong specification. A rule extracted incorrectly but with high confidence passes the reviewer, and every downstream agent then faithfully verifies the survey against a specification that is wrong.	The worst failure mode in the system, because nothing signals it. The pipeline works perfectly and returns a green report on a defective survey with an outcome strictly worse than no testing, since it confers false assurance.	Confidence derived from multi-sample agreement rather than model self-report, calibrated against hand-coded ground truth at M2 before Construction is authorized.
Dependency on Escalent. The project now relies on the sponsor for the QREs, for Decipher access and for the survey build itself. Any of the three arriving late stalls work that cannot start without it.	Schedule rather than quality, but concentrated at the front of the project where there is least slack. A late start on the evaluation set compresses every downstream gate.	Access and a first QRE set are confirmed as M1 exit criteria in Week 1. Until they land, agents are built against a stub Decipher adapter and a team-written placeholder QRE, so no agent work is blocked. Any slippage is raised in that week's progress email.

Success Metrics

These ten metrics confirm that what the platform produced was any good. They are fixed here, before any result exists. Each states its own denominator, so the table can be completed from the run ledger.

Metric	Target	How it is calculated
Programme deadlines met	All three dates	**Deadlines met / 3 : presentation 19 Oct, report 25 Oct, handover 30 Oct.** Binary per date.
Questionnaire read correctly	>95%	**Logic accuracy = semantically correct logic clauses / total logic clauses in the ground-truth specification.** Logic clauses are display predicates, validations, quotas, randomization rules, piping rules and terminations. Clauses marked ambiguous in ground truth score correct if the agent flagged them rather than resolved them. Reported as the mean over a QRE corpus on which the model is tested.
Escape Rate	<5%	**Escape rate = errors absent from the agent's open-issues queue / total errors found by the extraction comparison above.**
Testing time per QRE	Under 1 hour, against 8 to 16 hours today	**Total elapsed time from QRE upload to QC report download, including the reviewer's approval time.** **Measured on a client QRE of representative length and compared against the sponsor's stated baseline of 8 to 16 hours of manual testing per QRE.**

Metric	Target	How it is calculated
Survey logic tested	>90% of reachable paths	**Edge coverage = distinct graph edges traversed at least once / feasible edges.** **Feasible edges are those whose path condition the constraint solver finds satisfiable. Edges proven unsatisfiable are excluded from the denominator and enumerated separately in the QC report.**
Bugs found	At least 90% found, at most 10% false alarms	**Recall = seeded defects detected / seeded defects injected.** **False-alarm rate = reported defects adjudicated as not genuine / total defects reported.**
Same answer every time	100%	**Identical verdicts on re-adjudication / total verdicts.** The same captured artifacts are re-scored against the same approved specification. Any deviation is an infrastructure defect, since adjudication is deterministic code by design.
Time from upload to report	Under 45 minutes	**Sum of stage durations in the run ledger for ingestion, extraction, route planning, bot execution, adjudication and rendering, excluding the approval-gate time.** **Measured on a 60-question study at the declared worker count. This is machine time; the metric above is the end-to-end figure including human review.**
Sponsor can run it alone	Pass	**Binary, observed at UAT.** The sponsor uploads a QRE, approves the specification, starts a run and downloads the report with no team member intervening.

Progress Updates

The student's team will send update emails every week to the sponsor, faculty mentor and AMPBA program office. Each update will contain a summary of progress during the time and key insights that have been revealed, together with decisions taken, open risks and any asks for the week ahead.

Interaction	Date	Participants	Content	Owner
Weekly progress email	Every Sunday, from 16 Aug	Sponsor, faculty mentor, AMPBA programme office	Progress against plan, decisions taken, risks, asks for the week ahead	Anoop Kumar
Gate review M1 (Lifecycle Objectives)	Fri 14 Aug	Sponsor and team	Scope, success measures, working QRE set, Decipher access and automation feasibility	Atishi, Anoop
Gate review M2 (Lifecycle Architecture)	Fri 11 Sep	Sponsor and team	The full pipeline running end to end on one client QRE against a Decipher survey	Anoop, Anmol
Gate review M3 (Initial Operational Capability)	Fri 9 Oct	Sponsor and team	Every agent at its accuracy target; the full working QRE set passing end to end	Anmol, Raveena
Gate review M4 (Product Release) and UAT	Fri 16 Oct	Sponsor and team	Soft launch review, seeded-defect benchmark results, UAT sign-off	Sanskar, Atishi
Final presentation to sponsor	Mon 19 Oct	Sponsor, faculty mentor	End-to-end demonstration and outcomes against this charter	All members
Final report submission	Sun 25 Oct	AMPBA programme office	Final written report	Anoop, Raveena
Project handover	Fri 30 Oct	Sponsor	Code, documentation, environment, benchmark harness	All members

Other key milestones

Project Charter Finalization	7 August 2026
Weekly Reports	Every Sunday, commencing 16 August 2026
Final Presentation to Sponsor	By 19 October 2026
Final Report Submission	25 October 2026
Project Handover to the Sponsor	By 30 October 2026

Escalation and change control

Day-to-day queries are raised by the project manager with the client mentor by email, with a target response within three working days. Any matter unresolved for five working days is escalated to the faculty mentor. Any change to scope, deliverables or schedule will be raised through the Capstone Scope Change Request at the end of this charter, submitted by the sponsor to ISB and approved before any work changes.

Agreement

This charter and accompanying deliverables have been mutually discussed and agreed by:

Project Sponsor	Sameer Saurabh	
Team Member 1	Anmol Jindal	ANMOL
Team Member 2	Anoop Kumar	
Team Member 3	Atishi	Atishi
Team Member 4	Raveena Mallina	
Team Member 5	Sanskar Jain	

Capstone Scope Change Request

To be completed only if a change to scope, deliverables or schedule becomes necessary. The request is raised by the sponsor and submitted to ISB; no change takes effect until approved.

Name		Email Address	
Company	Escalent India Pvt. Ltd.	Telephone Number(s)	

Please describe the nature of the change requested:
Approximately how many additional hours of effort will this change require?
What other delays, costs, or risks would this change introduce?
What part of remaining project work can be forgone to accommodate this change?

Please email this form to hemanth_kumar@isb.edu. A decision will be conveyed within one week.

Student Team Recommendation:
Faculty Recommendations:
Other Considerations:
Accepted ☐ Rejected ☐